Dr. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY

LUCKNOW, UTTAR PRADESH

EVALUATION SCHEME & SYLLABUS

B. TECH. THIRD YEAR

BIOINFORMATICS

(As per AICTE Model Curriculum & NEP 2020)

Effective from the Session: 2026-27

SEMESTER - V												
Sr. No.	**Subject Code**	**Subject**	**Periods**			**Evaluation Scheme**			**End Sem.**		**Total**	**Credit**
			L	**T**	**P**	**CT**	**TA**	**Total**	**TE**	**PE**		
1	BBI501	Recombinant DNA Technology	3	1	0	20	10	30	70		100	4
2	BBI502	Fundamentals of Omics	3	1	0	20	10	30	70		100	4
3	BBI503	R Programming for Bioinformatics	3	1	0	20	10	30	70		100	4
4	BBI051-BBI054	Departmental Elective-I	3	0	0	20	10	30	70		100	3
5	BBI055-BBI058	Departmental Elective-II	3	0	0	20	10	30	70		100	3
6	BBI551	Recombinant DNA Technology Lab	0	0	2	20	10	30		50	100	1
7	BBI552	Fundamentals of Omics Lab	0	0	2			50		50	100	1
8	BBI553	R Programming for Bioinformatics Lab	0	0	2			50		50	100	1
9	BCC551	Mini Project or Internship Assessment*	0	0	2	20	10	30		50	100	1
10	BNC501/ BNC502	Constitution of India / Essence of Indian Traditional Knowledge	2	0	0	20	10	30	70			NC
11		MOOCs (Essential for Hons. Degree)										
		Total	**17**	**3**	**8**						**1000**	**22**

*The Mini Project or Internship (4 weeks) conducted during summer break after IV semester and will be assessed during V semester.

*It is desirable that the students should do their Summer Internship or Mini Project in their specialization area in line with the B.Tech. program

DEPARTMENTAL ELECTIVE - I	
Subject Code	**Subject**
BBI051	Principles of Cellular Biology
BBI052	Molecular Phylogeny and Evolution
BBI053	Metabolomics and Metabolic Engineering
BBI054	Fundamentals of Data Structures
DEPARTMENTAL ELECTIVE - II	
Subject Code	**Subject**
BBI055	Biophysics
BBI056	Internet Programming
BBI057	Pharmacogenomics and Pharmacogenetics
BBI058	Cancer Biology

SEMESTER - VI												
Sr. No.	**Subject Code**	**Subject**	**Periods**			**Evaluation Scheme**			**End Sem.**		**Total**	**Credit**
			L	**T**	**P**	**CT**	**TA**	**Total**	**TE**	**PE**		
1	BBI601	Immunoinformatics	3	1	0	20	10	30	70		100	4
2	BBI602	Artificial Intelligence and Machine Learning in Bioinformatics	3	1	0	20	10	30	70		100	4
3	BBI603	JAVA Programming	3	1	0	20	10	30	70		100	4
4	BBI061-BBI064	Departmental Elective-III	3	0	0	20	10	30	70		100	3
5	BOE060-BOE069	Open Elective-I	3	0	0	20	10	30	70		100	3
6	BBI651	Immunoinformatics Lab	0	0	2	20	30	50		50	100	1
7	BBI652	Artificial Intelligence and Machine Learning in Bioinformatics Lab	0	0	2	20	30	50		50	100	1
8	BBI653	JAVA Programming Lab	0	0	2	20	30	50		50	100	1
9	BNC601/ BNC602	Constitution of India / Essence of Indian Traditional Knowledge	2	0	0	20	10	30	70			NC
10		MOOCs (Essential for Hons. Degree)										
		Total	**17**	**3**	**6**						**800**	**21**

DEPARTMENTAL ELECTIVE - III	
Subject Code	**Subject**
BBI061	Gene Therapy
BBI062	Bioprocess Engineering
BBI063	Nano-Biosciences & Technology
BBI064	Biostatistics

B. TECH. (BIOINFORMATICS) — FIFTH SEMESTER DETAILED SYLLABUS

BBI501 RECOMBINANT DNA TECHNOLOGY		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO 1	Understand vectors and insert preparation (plasmids, phages, cosmids, BAC, YAC; restriction digestion, PCR).	**K2**
CO 2	Apply cloning techniques (ligation, modification enzymes, transformation, transfection, and screening).	**K3**
CO 3	Construct gene libraries (cDNA, subtractive cDNA, genomic, BAC, YAC).	**K3**
CO 4	Apply methods for expression of recombinant proteins in E. coli (expression vectors, promoters, reading frame, purification tags).	**K3**
CO 5	Understand the applications of DNA technology in diagnostics, therapeutics, agriculture, and genome editing, including AI-based rapid diagnostic technologies.	**K2**

DETAILED SYLLABUS		**L-T-P: 3-1-0**
Unit	**Topic**	**Proposed Lecture**
I	**VECTORS AND INSERT:** Vehicles for gene cloning - Biology of plasmids, Plasmid based vectors, Bacteriophage, Phage vectors, Cosmids, Phasmids, BAC and YAC vectors; Insert preparation-Restriction Digestion-Restriction enzymes, Polymerase chain reaction –Polymerase, Creation of restriction site by PCR.	08
II	**CLONING TECHNIQUE:** Ligation –Ligase, cohesive, Blunt end ligation; Other Modification enzymes – Nucleases, Phasphatase, Kinase; Transformation-Preparation of competent cell, chemical and electrical transformation; Transfection; Recombinant selection and screening -Blotting technique.	08
III	**CONSTRUCTION OF GENE LIBRARIES:** Construction of cDNA library- construction subtractive cDNA library – construction of genomic DNA library – BAC library – YAC library.	08
IV	**EXPRESSION OF RECOMBINANT PROTEIN IN E. COLI:** Plasmid expression vectors-general features, promoters used in expression vectors -cloning of genes in correct reading frame in expression vector- purification of recombinant protein using Histidine tag, GST tag, chitin.	08
V	**APPLICATION OF DNA TECHNOLOGY:** Applications of rDNA technology in Diagnostics and Therapeutics - HIV diagnosis, Human Insulin, Growth hormone; Recombinant Vaccines, Agricultural Technologies - Golden rice, Insect resistant crops, Genome Editing Tools using CRISPR Cas9 – Principle mechanism application ethics; Artificial Intelligence Application in RDT.	08

TEXT BOOKS / REFERENCES

1. Jerney D. Wale and Malcolm Von Schants, "From Genes to Genomes: Concepts and Applications of DNA Technology", John Wiley & Sons, 2002.
2. Sandy B. Primrose, Richard M. Twyman, "Principles of gene manipulation", 7th edition, Blackwell Publishing, 2006.
3. T A Brown, Gene cloning and DNA analysis, 6th edition, Blackwell Publishing, 2010.

BBI502 FUNDAMENTALS OF OMICS		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Explain foundational Omics concepts and the role of AI in biological data analysis.	**K2**
CO2	Apply AI tools for gene annotation of genome-sequencing data.	**K3**
CO3	Implement RNA-Seq pipelines and predict differential gene expression using AI.	**K3**
CO4	Explain protein identification methods and AI models for structure prediction.	**K2**
CO5	Apply AI-based integration of multiomics datasets for personalized healthcare solutions.	**K3**

DETAILED SYLLABUS		**L-T-P: 3-1-0**
Unit	**Topic**	**Proposed Lecture**
I	**INTRODUCTION TO OMICS:** Overview of genomics, transcriptomics, and proteomics; role of AI in biological data analysis and Omics.	08
II	**GENOMICS:** Genome sequencing, gene prediction, variant calling; AI tools for genome annotation (DeepVariant, AUGUSTUS, GENSCAN).	08
III	**TRANSCRIPTOMICS:** RNA-Seq data analysis, expression profiling, clustering and classification differential gene expression prediction. Role of AI in Transcriptomics.	08
IV	**PROTEOMICS:** Protein identification, quantitation, and structure prediction. Role of AI in Proteomics.	08
V	**MULTIOMICS DATA INTEGRATION:** Multiomics – Combination of data from genomics, proteomics, transcriptomics and metabolomics, AI in healthcare, diagnostics and personalized medicine.	08

TEXT BOOKS / REFERENCES

1. Andreas Baxevanis & B.F. Ouellette, Bioinformatics: A Practical Guide to the Analysis of Genes and Proteins, Wiley.
2. S.C. Rastogi et al., Bioinformatics: Methods and Applications, Springer.
3. Stuart Russell & Peter Norvig, Artificial Intelligence: A Modern Approach, Prentice Hall.
4. David Mount, Bioinformatics: Sequence and Genome Analysis, Cold Spring Harbor Laboratory Press.
5. S.C. Rastogi, Namita Mendiratta, Parag Rastogi, Bioinformatics: Methods and Applications – Genomics, Proteomics and Drug Discovery.

6. Singh, Bioinformatics: Genomics and Proteomics.
7. Devarajan Thangadurai, Jeyabalan Sangeetha, Genomics and Proteomics.

BBI503 R PROGRAMMING FOR BIOINFORMATICS		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Recall and explain the basics of R programming, environment setup, and representation of biological sequences.	**K2**
CO2	Apply conditional and looping constructs to develop simple biological programs (e.g., nucleotide counting, GC content).	**K3**
CO3	Use data structures and functions to handle biological datasets and perform string operations.	**K3**
CO4	Apply file handling techniques to import/export biological data and analyze FASTA sequences.	**K3**
CO5	Apply bioinformatics packages, statistics, visualization, and basic machine learning techniques for biological data analysis.	**K3**

DETAILED SYLLABUS		**L-T-P: 3-1-0**
Unit	**Topic**	**Proposed Lecture**
I	**INTRODUCTION TO R:** Introduction to R and its applications in bioinformatics, installation of R and RStudio, R environment, variables, constants, data types, operators, and basic R programs. Representation of DNA, RNA, and protein sequences.	08
II	**PROGRAM FLOW CONTROL:** Conditional statements (if, if-else), looping statements (for, while), break and next statements. Simple biological programs such as nucleotide counting and GC content calculation.	08
III	**DATA STRUCTURES AND FUNCTIONS:** Vectors, matrices, lists, data frames, factors, string operations, built-in functions, user-defined functions, and handling biological datasets.	08
IV	**FILE HANDLING:** Reading and writing text and CSV files, importing biological data, working with FASTA files, and simple sequence analysis.	08
V	**PACKAGES, STATISTICS AND VISUALIZATION:** Introduction to packages like ggplot2 and Bioconductor, basic statistics, graphs, charts, and introduction to clustering and classification Machine Learning algorithms in bioinformatics.	08

TEXT BOOKS / REFERENCES

1. Grolemund, G. (2014). Hands-On Programming with R. O'Reilly Media.
2. Wickham, H., & Grolemund, G. (2017). R for Data Science: Import, Tidy, Transform, Visualize, and Model Data. O'Reilly Media.
3. Matloff, N. (2011). The Art of R Programming: A Tour of Statistical Software Design. No Starch Press.

4. Peng, R. D. (2016). R Programming for Data Science. Leanpub.

BBI051 PRINCIPLES OF CELLULAR BIOLOGY		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO 1	Understand the origin, evolution, and molecular composition of cells, including enzymes, metabolism, and membranes.	**K2**
CO 2	Apply knowledge of cell organelles and bioenergetics (nucleus, ER, Golgi, mitochondria, chloroplasts, peroxisomes).	**K3**
CO 3	Explain the cytoskeleton, cell movement, transport, and cell–cell interactions.	**K2**
CO 4	Explain cell signaling, regulation, the cell cycle, programmed cell death, and stem cell applications.	**K2**
CO 5	Describe cellular diseases including cancer and neurodegenerative disorders.	**K2**

DETAILED SYLLABUS		**L-T-P: 3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**AN OVERVIEW OF CELLS AND CELL RESEARCH:** Origin and evolution of cells, cells as experimental models, tools of cell biology – chemistry of cells–molecular composition of cells, central role of enzymes, metabolic energy, biosynthesis of cell constituents, cell membrane.	08
II	**CELL STRUCTURE AND FUNCTION–I:** Nucleus, Endoplasmic reticulum, Golgi apparatus and Lysosomes, Bioenergetics and Metabolism – Mitochondria, chloroplasts, Peroxisomes.	08
III	**CELL STRUCTURE AND FUNCTION–II:** The cytoskeleton and cell movement, cell surface – transport of small molecules, Endocytosis, cell–cell interactions-Adhesion junctions-Tight junctions-Gap junctions-Plasmodesmata.	08
IV	**CELL SIGNALING–CELL REGULATION:** Signaling molecules and their receptors, functions, pathways of intracellular signal transduction–the Cell Cycle–Mitosis and Meiosis–Cell death and cell renewal- Programmed cell death-Stem cells-Embryonic stem cells and therapeutic cloning.	08
V	**DISEASES OF CELLS:** Epithelial cells and Cancer–Neurobiology and Neurodegenerative diseases.	08

TEXT BOOKS / REFERENCES

1. Geoffrey M. Cooper and Robert E. Hausman, “The Cell: A Molecular Approach”, Fifth Edition, ASM Press and Sinauer Associates, 2009.
2. Channarayappa, “Cell biology”, Orient Blackswan, 2010.
3. Rastogi SC, “Cell biology”, New Age International, 2005.

4. Cecie Starr, Ralph Taggart, "Biology: The Unity and Diversity of Life", Brooks/Cole, 11th Edition, 2006.

BBI052 MOLECULAR PHYLOGENY AND EVOLUTION		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Understand phylogenetic trees, their types, topological differences, and tree-building methods (UPGMA, LS, NJ, reconstruction).	**K2**
CO2	Apply maximum parsimony and maximum likelihood methods for consensus trees, branch length estimation, and protein data analysis.	**K3**
CO3	Explain statistical approaches in phylogenetics, including optimization principles, bootstrap tests, molecular clocks, and linearized trees.	**K2**
CO4	Apply parsimony and Bayesian approaches to infer ancestral sequences and explain synonymous/non-synonymous substitutions, convergent and parallel evolution.	**K3**
CO5	Explain the molecular basis of evolution, genetic polymorphism, and population trees derived from genetic markers.	**K2**

DETAILED SYLLABUS		**L-T-P: 3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**PHYLOGENETIC TREES:** Types- topological differences- tree building methods- Distance methods- UPGMA- LS methods- minimum evaluation methods- NJ methods- phylogenetic reconstruction.	08
II	**METHODS:** Maximum parsimony methods- Strategies of searching for MP trees- consensus trees- branch length estimation - weighted parsimony- MP methods for protein data- Maximum likelihood methods.	08
III	**STATISTICAL APPROACH:** Optimization principle and topological errors- interior branch tests- bootstrap tests- Tests of topological differences- advantages and disadvantages- molecular clocks and linearized trees.	08
IV	**ANCESTRAL SEQUENCES:** Inferences of ancestral sequences- parsimony and Bayesian approaches- synonymous and non-synonymous substitutions- convergent and parallel evolution.	08
V	**EVOLUTION:** Molecular basis of evolution- synonymous and non-synonymous mutations- genetic polymorphism and evolution- Population trees from genetic marker.	08

TEXT BOOKS / REFERENCES

1. Bruce S. Lieberman, "Phylogenetics: Theory and Practice of Phylogenetic Systematics", Wiley-Blackwell, second edition, 2011.
2. Masatoshi Nei, "Molecular Evolution and Phylogenetics", Oxford University Press, 2000.
3. David W Mount, "Bioinformatics- Sequence and genome analysis", Cold Spring Harbor Laboratory Press, second edition, 2004.
4. Roderick D.M. Page, Dr Edward C. Holmes, "Molecular Evolution: A Phylogenetic Approach", Blackwell Publishing, 1998.
5. Jin Xiong, "Essential Bioinformatics", Cambridge University Press, 2006.

BBI053 METABOLOMICS AND METABOLIC ENGINEERING		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Understand the role of metabolomics, analytical methods, databases, and reconstruction of metabolic networks.	**K2**
CO2	Apply bioinformatics tools and computational methods in metabolomics, including GC-MS pipelines and medical applications.	**K3**
CO3	Apply metabolic engineering principles to construct metabolic pathways.	**K3**
CO4	Apply metabolic flux analysis using isotope labeling, control analysis, and pathway grouping.	**K3**
CO5	Explain metabolic networks, kinetic models, and systems metabolic engineering approaches.	**K2**

DETAILED SYLLABUS		**L-T-P: 3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**INTRODUCTION TO METABOLOMICS:** Role of metabolomics in systems biology –application of metabolomics- Analytical methods in metabolomics – Data standards– Databases for Chemical, Spectral and Biological Data –Reconstruction of dynamic metabolic network model- examples- study of metabolome of a simple organism like E. Coli.	08
II	**BIOINFORMATICS IN METABOLOMICS:** Online databases and pipelines for metabolomics – GC-MS based metabolomics – Computational methods to compute and integrate metabolic data software for metabolomics- metabolomics and medical sciences.	08
III	**INTRODUCTION TO METABOLIC ENGINEERING:** Importance of metabolic engineering- comprehensive models for cellular reactions-material balances & data consistency- metabolic pathway synthesis.	08
IV	**METABOLIC FLUX ANALYSIS AND ITS APPLICATION:** Theory-determination of flux by isotope labeling-Metabolic control analysis- (control coefficients and summation theorems, FCC determination)-Grouping of reactions (gFCC, identification of independent pathways).	08
V	**METABOLIC NETWORKS:** Kinetic model of metabolic networks-Systems metabolic engineering of E. coli- bottom up and top down approaches of network analysis.	08

TEXT BOOKS / REFERENCES

1. Silas G. Villas-Boas, Jens Nielsen, Jorn Smedsgaard, Michael A. E. Hansen, Ute Roessner-Tunali, "Metabolome Analysis: An Introduction", John Wiley & Sons, 2007.
2. Jens Høiriis Nielsen, Michael C. Jewett, "Metabolomics: A Powerful Tool in Systems Biology", Springer, 2007.
3. Dr. Christoph Wittmann, Sang Yup Lee, "Systems Metabolic Engineering", Springer, 2012.
4. Gregory N. Stephanopoulos, "Metabolic Engineering- Principles and Methodologies", Academic Press, First Edition, 1998.
5. Ute Roessner, "Metabolomics", InTech, 2012.
6. Wolfram Weckwerth, "Metabolomics: Methods and Protocols", Humana Press, 2007.
7. Cortassa S., "An Introduction to Metabolic and Cellular Engineering", World Scientific Publishing Company Ltd., 2002.

BBI054 FUNDAMENTALS OF DATA STRUCTURES		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Explain linear data structures (Lists, Stacks, Queues) and their array and linked implementations with asymptotic runtime complexity.	**K2**
CO2	Apply and implement balanced non-linear data structures (AVL, Splay, B-Trees) and priority queues to solve hierarchical data problems.	**K3**
CO3	Construct and apply efficient sorting algorithms (such as Quick Sort, Merge Sort, and Heap Sort) based on time and space metrics.	**K3**
CO4	Apply graph-based solutions using topological sorting, shortest-path algorithms, and minimum spanning trees for networks.	**K3**
CO5	Apply advanced tree searching techniques and hashing strategies for efficient data retrieval.	**K3**

DETAILED SYLLABUS		**L-T-P: 3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**LINEAR DATA STRUCTURES:** Algorithm Analysis - Mathematical background, Model - Running Time Calculations - Linear Data Structure - List ADT Array and Cursor Implementations - Stack ADT - Queue ADT Array and List Implementation and Application.	08
II	**NON-LINEAR DATA STRUCTURES:** Trees- Binary trees - Binary Tree Representations, Tree Traversals, AVL Trees- Single and Double Rotation, Splay trees- B-Trees- Priority Queue- Binary Heap, Applications and variations of Priority Queue.	08
III	**SORTING METHODS:** Sorting- Exchange sorts- Selection and Tree sorting- Insertion Sort- Heap Sort- Merge Sort- Quick Sort- Bucket Sort.	08
IV	**GRAPH ALGORITHMS:** Definition and Representation of graphs- Topological sort- Shortest-Path Algorithms- Network flow problems- Minimum Spanning Tree- Application of Depth-First Search.	08
V	**SEARCHING:** Basic Search Techniques- Tree Searching- General Search Trees- Hashing.	08

TEXT BOOKS / REFERENCES

1. Jeffrey Esakov, Tom Weiss, "Data Structures An Advanced Approach using C", PHI, 1989.
2. Mark Allen Weiss, "Data Structures and Algorithm Analysis in C", Addison Wesley Publications, 2007.
3. Aaron M. Tanenbaum, Yedidyah Langsam, Moshe J. Augensten, "Data Structures Using C", PHI, 1990.
4. Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman, "Data Structures and Algorithms", Pearson Education, 10th Reprint, 2004.
5. Jean-Paul Tremblay, Paul G. Sorenson, "An Introduction to Data Structures with Applications", 2nd Edition, Tata McGraw-Hill Edition, 1976.

BBI055 BIOPHYSICS		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Understand fundamental principles of biophysics including atomic interactions, mechanics, and thermodynamics.	**K2**
CO2	Apply DNA biophysics concepts such as structural forms, torsion angles, and helical parameters.	**K3**
CO3	Explain protein biophysics including peptide bonds, Ramachandran plots, and biomolecular interactions.	**K2**
CO4	Explain carbohydrate biophysics covering monosaccharides, polysaccharides, and conjugates.	**K2**
CO5	Apply X-ray crystallography, NMR, spectroscopy, Mass Spectrometry, and MALDI-TOF techniques to study biomolecules.	**K3**

DETAILED SYLLABUS		**L-T-P: 3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**ELEMENTARY BIOPHYSICS RESEARCH:** Definition of atom and its properties, size and charge of electron, nucleus, protons & neutrons. Types of interaction between atoms: Bond distance, Bond angle, Torsion Angle, H-bonding, VdW interactions, Hydrophobic interactions, Electrostatics interactions. Classical Mechanics: Newton's laws of motion, Law of conservation of mass and energy. Thermodynamics: Laws, Enthalpy, Entropy, Free Energy, Gibbs & Helmholtz free energy, Internal energy.	08
II	**DNA BIOPHYSICS:** Structure of A, B & Z forms of DNA with special emphasize to formation of phosphodiester bond, hydrogen bonding between the two strands of DNA, Helical parameters, DNA Backbone torsion angles, sugar back bone torsion angles, pseudo rotation, sugar puckering. Comparison of various parameters of A, B & Z forms of DNA.	08
III	**PROTEIN BIOPHYSICS:** Amino acids, peptide bond, phi, psi & omega angles with distance, Zwitterion formation of amino acids, Levels of protein structure, Significance of Ramachandran Plot, Protein – DNA, drug, carbohydrate, small molecule interactions. Energetics of lipid bilayer. Molecular Mechanics & Dynamics protocol.	08
IV	**CARBOHYDRATE BIOPHYSICS:** Structure of D-glucose & D-fructose; formation of glucosides & the cyclic structure of D-glucose; D-ribose & D-deoxyribose; Structure and conformation of disaccharides and polysaccharides-cellulose, amylose, amylopectin & glycogen, Chitin, carbohydrate conjugates.	08

V	**TECHNIQUES IN BIOPHYSICS:** X-ray crystallography, NMR, IR, Raman, AAS, UV, CD, Cosy, Noesy, spectroscopy, Mass spectrometry, Maldi TOF for protein prediction.	08

TEXT BOOKS / REFERENCES

1. Carl Branden & John Tooze (1999), "Introduction to Protein Structure", Garland Publishing, New York & London.
2. Wolfram Saenger (1983), "Principles of Nucleic Acid Structure", Springer-Verlag, New York.
3. Andrew R. Leach (2000), "Molecular Modelling Principles and Applications", Prentice Hall.
4. Puri B.R., L.R. Sharma, M.S. Pathania (2008), "Principles of Physical Chemistry", Vishal Publishing Company.
5. Murugesan R. (2004), "Modern Physics", S. Chand & Co.
6. Vasantha Pattabhi & Gautham (2002), "Biophysics", Narosa Publishing.

BBI056 INTERNET PROGRAMMING		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Explain the basics of the internet and HTML.	**K2**
CO2	Apply scripting languages for web-based tasks.	**K3**
CO3	Use markup languages XML and XHTML for structuring web data.	**K3**
CO4	Apply server-side programming concepts using servlets and JDBC.	**K3**
CO5	Apply the fundamentals of ASP.NET to develop web-based applications.	**K3**

DETAILED SYLLABUS		**L-T-P: 3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**INTERNET BASICS, HTML:** Basic internet concepts - web server, client – Apache server – Internet Information Services (IIS)- Markup Language- Introduction to HTML and Cascading Style Sheets (CSS)- Internet applications- Email – chat – search engines – news groups.	08
II	**SCRIPTING LANGUAGES:** Java Scripting - VB scripting- Object model and event model- Document Object Model (DOM) – Common Gateway Interface (CGI) & data base connectivity- Introduction to PHP and Perl scripting.	08
III	**XML AND XHTML:** Introduction to XML – Document Type Definitions - XML schemas – XML Database creation. Introduction to XHTML and XHTML elements.	08
IV	**SERVER SIDE PROGRAMMING:** Multitier application- Introduction to Java servlets- HTTP GET & POST request- JDBC principles- cookies session tracking Principal Components Regression- Partial Least Squares- Molecular Field Analysis.	08
V	**ASP.NET:** Introduction to ASP.NET- using objects- data types- data access objects- connection object- command object. CASE STUDY- Developing a web based tool for sequence analysis.	08

TEXT BOOKS / REFERENCES

1. Steven M. Schafer, “HTML, XHTML, and CSS Bible”, Fifth Edition, Wiley Publishing, Inc., 2010.
2. Deitel and Deitel, Goldberg, “Internet & World Wide Web, How To Program”, third edition, Pearson Education, 2004.
3. Paul Wilton and Jeremy McPeak, “Beginning JavaScript”, Wrox, 4th edition, 2009.

4. Cerami Ethan, "XML for Bioinformatics", Springer, 2005.
5. Adrian Kingsley-Hughes, Kathie Kingsley-Hughes, Daniel Read, "VBScript Programmer's Reference", 3rd Edition, Wrox, 2007.
6. Vivek Chopra, Jon Eaves, Rupert Jones, Sing Li, John T. Bell, "Beginning JavaServer Pages", Wrox, 2005.
7. Macdonald Mathew, "ASP.NET - The Complete Reference", McGraw-Hill/Osborne, 2002.

BBI057 PHARMACOGENOMICS & PHARMACOGENETICS		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Understand historical aspects of pharmacogenetics, pharmacogenomics, biomarkers, and personalized medicine.	**K2**
CO2	Apply knowledge of pharmacogenetics in drug metabolism, receptors, and transporters.	**K3**
CO3	Explain inter-ethnic drug responses, clinical viewpoints, and related technologies.	**K2**
CO4	Explain single nucleotide polymorphisms (SNPs) and technologies for molecular diagnostics.	**K2**
CO5	Explain Serial Analysis of Gene Expression (SAGE) for functional biology and disease mapping.	**K2**

DETAILED SYLLABUS		**L-T-P: 3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**INTRODUCTION:** Historical aspects of Pharmacogenetics- Pharmacogenomics- Biomarkers- and the promise of personalized medicine.	08
II	**PHARMACOGENETICS:** Pharmacogenetics of drug metabolism- receptors- drug transporters.	08
III	**DRUG RESPONSE:** Inter ethnic drug response- clinical viewpoints- technologies and challenges.	08
IV	**SINGLE NUCLEOTIDE POLYMORPHISM:** Introduction- technologies for the analysis of SNPs- molecular diagnostics.	08
V	**SAGE:** Serial Analysis of Gene Expression (SAGE) - Principle and methodology of SAGE; applications in functional biology; use of SAGE data for mapping of disease loci.	08

TEXT BOOKS / REFERENCES

1. Licinio J. Ma-Li Wong, “Pharmacogenomics: The Search for Individualized Therapies”, Wiley-VCH, 2002.
2. Zdanowicz M, “Concepts in Pharmacogenomics”, ASHP, 2010.
3. Werner Kalow, Rachel F Tyndale, Urs A Meyer, “Pharmacogenomics”, Marcel Dekker Inc., 2001.
4. Adam Hedgecoe, “The Politics of Personalized Medicine- Pharmacogenetics in the Clinic”, Cambridge University Press, first edition, 2004.

BBI058 CANCER BIOLOGY		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Understand fundamentals of cancer biology including cell cycle regulation, mutations, tumour suppressor genes, and cancer types.	**K2**
CO2	Apply principles of carcinogenesis (chemical, physical, radiation mechanisms).	**K3**
CO3	Explain molecular cell biology of cancer focusing on oncogenes, growth factors, and transformations.	**K2**
CO4	Explain principles of cancer metastasis including invasion, heterogeneity, and the metastatic cascade.	**K2**
CO5	Describe new molecules and therapies for cancer, including chemotherapy, radiation, bioinformatics, and advances in detection and AI integration.	**K2**

DETAILED SYLLABUS		**L-T-P: 3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**FUNDAMENTALS OF CANCER BIOLOGY:** Regulation of Cell cycle- Mutations that cause changes in signal molecules- effects on receptor- signal switches tumour suppressor genes- Modulation of cell cycle-in cancer- Different forms of cancers- Diet and cancer.	08
II	**PRINCIPLES OF CARCINOGENESIS:** Chemical Carcinogenesis- Metabolism of Carcinogenesis- Natural History of Carcinogenesis- Targets of Chemical Carcinogenesis- Principles of Physical Carcinogenesis- X-Ray radiation - Mechanism of radiation Carcinogenesis.	08
III	**PRINCIPLES OF MOLECULAR CELL BIOLOGY OF CANCER:** Oncogenes- Identification of Oncogenes- Retroviruses and Oncogenes- detection of Oncogenes- Growth factor and Growth factor receptors that are Oncogenes- Oncogenes/ Proto Oncogenes activity- Growth factors related to transformations.	08
IV	**PRINCIPLES OF CANCER METASTASIS:** Clinical significances of invasion- heterogeneity of metastatic phenotype- Metastatic cascade- Basement membrane disruption- Three step theory of invasion- Proteinases and tumour cell invasion.	08
V	**NEW MOLECULES FOR CANCER THERAPY:** Different forms of therapy- Chemotherapy- Radiation Therapy- Detection of Cancers- Prediction of aggressiveness of Cancer- Advances in Cancer detection- Bioinformatics and Cancer. Role of machine	08

	learning and deep learning in cancer research. Cancer screening and detection via AI integration.	

TEXT BOOKS / REFERENCES

1. King R. J. B., "Cancer Biology", Addison Wesley Longmann Ltd, U.K., 1996.
2. Ruddon R.W., "Cancer Biology", Oxford University Press, Oxford, 1995.
3. Maly B.W.J., "Virology a practical approach", IRL press, Oxford, 1987.
4. Dunmock N.J. and Primrose S.B., "Introduction to modern Virology", Well Scientific Publications, Oxford, 1988.

BBI551 RECOMBINANT DNA TECHNOLOGY LAB		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Apply restriction digestion, ligation, and PCR techniques for DNA manipulation.	**K3**
CO2	Prepare competent cells, perform transformation in E. coli, and examine recombinant clones.	**K3**
CO3	Isolate genomic and plasmid DNA, purify digested fragments, and confirm insert integration.	**K3**
CO4	Examine recombinant constructs using agarose gel electrophoresis, PCR products, and blue-white screening.	**K3**
CO5	Demonstrate practical skills in recombinant DNA technology workflows from DNA isolation to gene amplification.	**K3**

LIST OF EXPERIMENTS
1. Restriction enzyme digestion of plasmid **2.** Purification of digested DNA **3.** Ligation of DNA fragment with plasmid DNA **4.** Preparation of competent cells **5.** Transformation in E. coli **6.** Isolation of recombinant plasmid and confirmation of insert DNA **7.** Blue-white screening of transformants **8.** Polymerase Chain Reaction **9.** Sanger Sequencing **10.** Next Generation Sequencing
Note: The instructor may add / delete / modify / tune experiments as required.

BBI552 FUNDAMENTALS OF OMICS LAB		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Demonstrate knowledge of open biological databases and AI based tools.	**K2**
CO2	Retrieve and process genomic, transcriptomic, and proteomic data using web servers.	**K3**
CO3	Apply alignment, clustering, and prediction methods with AI support.	**K3**
CO4	Apply computational tools to identify protein structures, motifs, and gene expression patterns.	**K3**
CO5	Explain AI applications in healthcare and multi-omics integration.	**K2**

LIST OF EXPERIMENTS

1. Retrieve DNA sequences from NCBI and visualize basic sequence features.
2. Primer designing.
3. Perform gene prediction using AUGUSTUS (web server demo).
4. Explore mutation detection using DeepVariant demo dataset.
5. Load RNA-Seq sample data in Galaxy and generate expression plots.
6. Cluster transcriptomic data using Orange Data Mining (drag-and-drop).
7. Retrieve protein sequences from UniProt and analyze domains using InterPro.
8. Predict protein secondary structure using SOPMA/PSIPRED online tools.
9. Phosphorylation site prediction using NetPhos.
10. Peptide structural characterization by Bioinformatics tool.
11. Identify motifs in protein sequences using HMMER web server.
12. Map genes/proteins to pathways using KEGG Mapper.
13. Run a simple neural network in Google Colab to classify DNA vs. protein sequences.
14. Apply K-means clustering on demo gene expression data (Orange/Colab).
15. Case study: Use Orange's built-in healthcare dataset to train a decision tree classifier.

Note: The instructor may add / delete / modify / tune experiments as required.

BBI553 R PROGRAMMING FOR BIOINFORMATICS LAB		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Write simple R programs using variables, operators, and input/output statements for biological data representation.	**K3**
CO2	Implement conditional and looping constructs to solve biological problems such as nucleotide counting and GC content calculation.	**K3**
CO3	Create and manipulate data structures (vectors, matrices, lists, data frames) and develop user-defined functions for biological datasets.	**K3**
CO4	Perform file handling operations to read/write CSV and FASTA files and conduct basic sequence analysis.	**K3**
CO5	Apply statistical methods and visualization techniques (plots, graphs, clustering) to analyze and interpret biological data.	**K3**

LIST OF EXPERIMENTS

1. Write simple R programs using variables, operators, and basic input-output statements.
2. Implement programs using conditional statements (if, if-else) for simple biological decision-making problems.
3. Write programs using loops (for, while) for repetitive biological calculations.
4. Create and manipulate vectors and matrices for storing DNA sequences and biological data.
5. Implement programs using lists and data frames for organizing gene or sample information.
6. Write user-defined functions for simple biological operations.
7. Write a program to count nucleotide frequency (A, T, G, C) in a DNA sequence.
8. Write a program to calculate GC content and sequence length of a DNA sequence.
9. Read biological data from CSV/text files and perform basic data analysis.
10. Read and analyze sequence data from FASTA files.
11. Perform basic statistical analysis such as mean, median, standard deviation, and correlation on biological datasets.
12. Create graphical representations such as bar plots, histograms, scatter plots, and gene expression plots using R.

Note: The instructor may add / delete / modify / tune experiments as required.

B. TECH. (BIOINFORMATICS) — SIXTH SEMESTER DETAILED SYLLABUS

BBI601 IMMUNOINFORMATICS		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Understand sequence analysis methods including DNA alignments, molecular evolution, viral evolution, and prediction of functions.	**K2**
CO2	Apply computational methods in immunological bioinformatics (sequence weighting, clustering, Gibbs Sampling, HMM, neural networks, microarray applications).	**K3**
CO3	Apply prediction methods for cytotoxic T cell (MHC Class I) epitopes and explain antigen processing pathways.	**K3**
CO4	Apply prediction methods for helper T cell (MHC Class II) epitopes and describe their processing mechanisms.	**K3**
CO5	Apply B cell epitope prediction approaches and web-based tools for vaccine design.	**K3**

DETAILED SYLLABUS		**L-T-P: 3-1-0**
Unit	**Topic**	**Proposed Lecture**
I	**SEQUENCE ANALYSIS:** Alignments- DNA alignments- Molecular evolution and phylogeny- viral evolution and escape- prediction of functions.	08
II	**METHODS:** Methods applied in Immunological Bioinformatics- starting from sequence weighing methods to cluster analysis- Gibbs Sampling- HMM- Neural network- microarray and its applications.	08
III	**MHC-I PREDICTION:** Prediction of Cytotoxic T Cell (MHC Class I) Epitopes- Antigen Processing in the MHC Class I Pathway.	08
IV	**MHC-II PREDICTION:** Prediction of Helper T Cell (MHC Class II) Epitopes- Processing of MHC Class II Epitopes.	08
V	**B CELL EPITOPE PREDICTION AND WEB SOURCES:** Recognition of Antigen by B Cells- vaccine design- Web-Based Tools for Vaccine Design.	08

TEXT BOOKS / REFERENCES

1. Rammensee, "Immunoinformatics- Bioinformatics Strategies for Better Understanding of Immune Function", Wiley, 2003.
2. Ole Lund, "Immunological Bioinformatics", MIT Press, September 2005.
3. Darren Flower, "In Silico Immunology", Springer, 2006.

4. Darren R Flower, "Immunoinformatics: Predicting Immunogenicity in Silico", Humana Press, 2007.

BBI602 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING IN BIOINFORMATICS		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Explain the fundamental concepts of Artificial Intelligence.	**K2**
CO2	Describe modern AI technologies and their applications in intelligent systems and healthcare.	**K2**
CO3	Apply the fundamental concepts and workflow of machine learning for data analysis and predictive modeling.	**K3**
CO4	Apply supervised and unsupervised machine learning algorithms to solve classification, regression and clustering problems.	**K3**
CO5	Apply AI and machine learning approaches to solve bioinformatics and healthcare problems.	**K3**

DETAILED SYLLABUS		**L-T-P: 3-1-0**
Unit	**Topic**	**Proposed Lecture**
I	**FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE:** Introduction to Artificial Intelligence, history and evolution of AI, types of AI (Narrow, General and Super AI), intelligent agents, problem-solving approaches, knowledge representation and reasoning, search techniques (Breadth First Search, Depth First Search and A* Search).	08
II	**MODERN ARTIFICIAL INTELLIGENCE:** Expert systems, Natural Language Processing (NLP), computer vision, robotics, introduction to Generative AI and Large Language Models (LLMs), AI ethics, bias, privacy and security, and emerging trends in Artificial Intelligence.	08
III	**FUNDAMENTALS OF MACHINE LEARNING:** Introduction to Machine Learning, Artificial Intelligence vs Machine Learning vs Deep Learning, machine learning workflow, types of machine learning (Supervised Learning, Unsupervised Learning and Reinforcement Learning), data collection and pre-processing, feature selection, training, validation and testing datasets, overfitting and under fitting, and model evaluation using accuracy, precision, recall, F1-score and confusion matrix.	08
IV	**SUPERVISED AND UNSUPERVISED LEARNING ALGORITHMS:** Supervised learning algorithms: Linear Regression, Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbour (KNN) and Naïve Bayes. Unsupervised learning algorithms: K-Means Clustering, Hierarchical Clustering and Principal Component	08

	Analysis (PCA). Introduction to Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN).	
V	**APPLICATIONS OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING IN BIOINFORMATICS:** Applications of Artificial Intelligence and Machine Learning in genomics, proteomics, transcriptomics, biological sequence analysis, disease diagnosis, drug discovery, protein structure prediction, precision medicine, medical image analysis, personalized medicine, AI-assisted clinical decision support systems, and ethical considerations in AI-based healthcare.	08

TEXT BOOKS / REFERENCES

1. Stuart Russell, Peter Norvig, "Artificial Intelligence: A Modern Approach", 4th Edition, Pearson Education, 2021.
2. Aurélien Géron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow", 3rd Edition, O'Reilly Media, 2022.
3. Arthur Lesk, "Introduction to Bioinformatics", 5th Edition, Oxford University Press, 2019.
4. David W. Mount, "Bioinformatics: Sequence and Genome Analysis", 2nd Edition, Cold Spring Harbor Laboratory Press, 2004.

BBI603 JAVA PROGRAMMING		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Understand the basics of Java, object-oriented programming concepts, data types, variables, arrays, and operators.	**K2**
CO2	Apply control statements, create classes and methods, and implement inheritance concepts.	**K3**
CO3	Use packages, interfaces, exception handling, applets, and string handling techniques.	**K3**
CO4	Implement event handling and utilize AWT controls for GUI development.	**K3**
CO5	Apply Java and Java-based tools for bioinformatics analysis.	**K3**

DETAILED SYLLABUS		**L-T-P: 3-1-0**
Unit	**Topic**	**Proposed Lecture**
I	**INTRODUCTION TO JAVA:** An Overview of Java- Object Oriented Programming, A Simple JAVA Program. Data Types - Variables, and Arrays, Operators– Arithmetic Operators, Bitwise Operators, Relational Operators and Logical Operators.	07
II	**CONTROL STATEMENTS, CLASSES, METHODS AND INHERITANCE:** Control Statements:- Selection statements, Iteration statements, Jump statements. Classes and Methods:- Introducing Classes, fundamentals, declaring objects, Introducing Methods, constructors, keywords. Inheritance:- Basics, 'super', hierarchy, method overriding, abstract and object class.	09
III	**PACKAGES, APPLETS AND STRING HANDLING:** Packages and Interfaces. Exception Handling:- fundamentals, types, 'try and catch', 'throw', 'finally', built-in exceptions. I/O Applets- the Applet class, String handling:- string constructors, operations, character extraction, string comparison, modifications and string buffer.	08
IV	**EVENT HANDLINGS AND AWT CONTROLS:** Event Handling- Key event class-Mouse event class- Text event Class- Event listener interfaces-Programs for handling mouse and keyboard events- Using AWT controls-Labels- Text field-Buttons-Check boxes- Choice controls and Scroll bars- Layout Managers and Menus.	09
V	**APPLICATIONS OF JAVA:** Applications of Java in Bioinformatics: Bio-JAVA-Java based software used for Bioinformatics analysis (Pattern Hunter)-Java code examples.	07

TEXT BOOKS / REFERENCES

1. Herbert Schildt, "The Complete Reference JAVA 2", Fifth Edition, Tata McGraw Hill, 2005.
2. Harvey M. Deitel, Paul J. Deitel, "Java: How to Program: with an Introduction to Visual J++", Prentice Hall, 1997.
3. Kim B. Bruce, Thomas P. Murtagh, Andrea Pohoreckyj Danyluk, "Java: An Eventful Approach", Prentice Hall College, 2006.
4. Joseph P. Russell, "Java Programming for the Absolute Beginner", Thomson Course Technology, 2001.

BBI061 GENE THERAPY		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Understand general principles of gene therapy, delivery methods, and clinical trial approaches.	**K2**
CO2	Apply DNA-based gene therapy techniques including mechanical and liposomal delivery.	**K3**
CO3	Explain retroviral and adenoviral vectors, their safety, and case studies (OTC, SCID).	**K2**
CO4	Explain adeno-associated virus vectors, tissue-specific promoters, and targeted delivery strategies.	**K2**
CO5	Describe cancer gene therapy methods including prodrug activation, oncolytic viruses, DNA vaccines, and antisense approaches.	**K2**

DETAILED SYLLABUS		**L-T-P: 3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**GENERAL PRINCIPLES OF GENE THERAPY:** Diseases that could be treated by gene therapy- Clinical trials- Gene Delivery. Reporter genes and Transfer efficiency- Germline vs. somatic cells- In vivo and ex vivo gene therapy- Ex vivo and In vivo Gene Therapy.	09
II	**DNA–BASED GENE THERAPY:** Mechanical methods of delivery- Example: Duchenne myotrophy- Liposomal methods of delivery- Cystic fibrosis.	07
III	**ADENO-ASSOCIATED VIRUS-BASED VECTORS:** Safety of GT vectors- Gutted vectors- Helper viruses and Packaging cell lines- Examples: ornithine transcarbamylase deficiency (OTC) - Severe Combined Immunodeficiency (SCID). -Failures in the OTC and SCID GT trials.	08
IV	**EVENT HANDLINGS AND AWT CONTROLS:** Hemophilia- Diabetes mellitus- Erythropoietin- How to target vector to the particular cell- Tissue specific promoters- Vectors as ligands for cellular receptors.	08
V	**CANCER GENE THERAPY:** Genetic Prodrug Activation Gene Therapy- GT methods to overcome chemotherapy resistance of tumors- Biological response modifiers- Oncolytic virus-based strategies of tumor treatment. DNA vaccines for cancer and infectious diseases- AIDS as a target for DNA vaccines- Antisense based methods of gene therapy.	08

TEXT BOOKS / REFERENCES

1. Peter J. Quesenberry, "Stem cells biology and gene therapy", Wiley-Liss, 1998.
2. Keith Greenberg, "Gene therapy", Blackbirch Pr Inc, 2003.

BBI062 BIOPROCESS ENGINEERING		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Explain the principles of bioprocess engineering, microbial growth, and fermentation systems.	**K2**
CO2	Apply bioreactor design concepts and scale-up criteria for industrial bioprocesses.	**K3**
CO3	Explain mixing, mass transfer, and oxygen transfer phenomena in bioreactors.	**K2**
CO4	Apply sterilization methods, strain improvement techniques, and media formulation strategies in fermentation processes.	**K3**
CO5	Explain immobilized cell systems and proteomic techniques for modern bioprocess and biotechnology applications.	**K2**

DETAILED SYLLABUS		**L-T-P: 3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**INTRODUCTION:** Introduction to Bioprocess, Microbial process development, Fermentation Economics, Batch culture, Continuous culture, kinetics, Chemostat with recycle and multistage Chemostat Systems. Modify batch reactors: Fed-batch operation, Perfusion systems.	08
II	**BIOREACTOR DESIGN SCALE UP, SCALE DOWN:** Bioreactor configurations, Analysis of ideal and non-ideal reactors. Multiphase reactors: packed-bed reactors, bubble-column bioreactors, fluidized bed bioreactors, trickle-bed reactors. Bioprocess Considerations: Animal cell cultures & plant cell cultures Scale up of bioprocesses-based on oxygen transfer, power consumption and impeller tip speed, Scale down.	09
III	**MIXING & MASS TRANSFER:** Mixing equipments, flow patterns in reactors, mixing mechanism, power consumption and shear properties of sparged and agitated vessels and various mixing agitators. Diffusion, mass transfer, Oxygen transfer in fermenters.	08
IV	**STERILIZATION, STRAIN IMPROVEMENT AND MEDIA FORMULATION:** Design of batch and continuous sterilization processes, kinetics of thermal death of cells and spores. Strain improvement of industrially important microorganisms, Media formulation industrial fermentations.	08
V	**IMMOBILIZED CELL SYSTEMS (ICS):** Mass Analysis-Tandem Mass Spectrometry (MS/MS)-Ion Detectors-Ionization of the Sample. Sample Preparation/Analysis-Proteins and Peptides-Protein Folding.	07

TEXT BOOKS / REFERENCES

1. Pauline M. Doran, "Bioprocess Engineering Principles", 8th ed., Academic Press, New York, 2003.
2. M.L. Shuler and F. Kargi, "Bioprocess Engineering--basic Concepts", 2nd Edn., Prentice-Hall Of India Pvt Ltd (2008).
3. Peter F. Stanbury, Stephen J. Hall & A. Whitaker, "Principles of Fermentation Technology", Elsevier India Pvt Ltd. (2007).
4. Klaas Van't Riet, Johannes Tramper, "Basic Bioreactor Design", 2nd ed., Marcel Dekker, Inc., New York, 1991.
5. Bailey and Ollis, "Biochemical Engineering Fundamentals", 2nd ed., McGraw-Hill Book Company, New York, 1986.
6. Mccabe L. Warren, Smith C. Julian and Peter Harriott, "Unit Operations of Chemical Engineering", 6th ed., McGraw Hill International Edition, New York, 2001.
7. Abhilasha S. Mathuriya, "Industrial Biotechnology", 1st ed., Ane Books Pvt. Ltd., New Delhi, 2009.

BBI063 NANO-BIOSCIENCES & TECHNOLOGY		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Understand the fundamental concepts, history, and approaches of nanobiotechnology, including micro and nanofabrication techniques.	**K2**
CO2	Explain the properties, synthesis, and applications of carbon nanotubes and metal nanoparticles (gold, silver, zinc oxide).	**K2**
CO3	Explain the principles and functioning of AFM and STM and their role in nanodiagnostics and drug delivery systems.	**K2**
CO4	Describe the synthesis and characterization of biomedical polymers and their applications in pharmaceutical and medical fields.	**K2**
CO5	Explain nano biosensors, imaging agents, and tumor-targeting technologies using quantum dots.	**K2**

DETAILED SYLLABUS		**L-T-P: 3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**INTRODUCTION TO NANOBIOTECHNOLOGY:** History, origin, and fundamental concepts. Bottom-up and top-down approaches. Micro and nanofabrication—discussion, current research trends, tools and techniques, applications, implications, and nanofabrication processes.	08
II	**CARBON NANOTUBES & RELATED STRUCTURES**—properties, synthesis, and applications. Types and synthesis of metal nanoparticles. Applications of gold, silver, and zinc oxide nanoparticles. Nano chemicals and their significance.	09
III	**NANOSCALE IMAGING AND DIAGNOSTIC TOOLS:** Atomic Force Microscopy (AFM) and Scanning Tunneling Microscopy (STM). Advanced nanodiagnostic devices and drug delivery tools developed through nanotechnology.	09
IV	**BIOMEDICAL POLYMERS:** Synthesis and characterization of biomedical polymers. Applications in pharmaceutical, cardiovascular, ophthalmologic, and orthopedic fields.	07
V	**BIO-NANOTECHNOLOGY & TUMOR TARGETING:** Micro and nano biosensors, bioavailability, nanoimaging agents, and tumor targeting through nanotechnology. Quantum dots technology and its biomedical applications.	07

TEXT BOOKS / REFERENCES
1. Mark Ratner and Daniel Ratner, “Nanotechnology”, Pearson Education.

2. Guozhong Cao, "Nanostructures and Nanomaterials, synthesis, properties and applications", Imperial College Press, 2004.
3. Hari Singh Nalwa, "Nanostructured Materials and Nanotechnology", Academic Press, 2002.
4. Mark James Jackson, "Microfabrication and Nanomanufacturing".
5. A.R. Jha, "MEMS and Nanotechnology–Based sensors and devices communication, Medical and Aerospace applications".
6. Subbiah Balaji, "Nano Biotechnology", MJP Publishers.

BBI064 BIOSTATISTICS		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Explain fundamental statistical concepts including data types, central tendency, and dispersion.	**K2**
CO2	Apply probability theory and probability distributions to model biological events.	**K3**
CO3	Apply statistical inference and hypothesis testing to biological data.	**K3**
CO4	Apply regression, survival analysis, and multivariate techniques to biological datasets.	**K3**
CO5	Use statistical software and data visualization tools to interpret biological data.	**K3**

DETAILED SYLLABUS		**L-T-P: 3-1-0**
Unit	**Topic**	**Proposed Lecture**
I	**FUNDAMENTAL STATISTICAL CONCEPTS:** Data Types: Nominal, Ordinal, Interval, Ratio; Measures of Central Tendency: Mean, Median, Mode; Measures of Dispersion: Range, Variance, Standard Deviation; Application in Bioinformatics: Central tendency and dispersion in gene expression data, clinical trials, etc.	08
II	**PROBABILITY AND PROBABILITY DISTRIBUTIONS:** Basic Probability Theory: Conditional probability, Bayes' theorem; Probability Distributions: Normal, Binomial, Poisson, and Exponential distributions; Applications: Using probability distributions in modeling biological events (e.g., mutation rates, genetic inheritance patterns).	08
III	**STATISTICAL INFERENCE AND HYPOTHESIS TESTING:** Hypothesis Testing Basics: Null and Alternative Hypothesis, Type I and Type II errors, p-value interpretation; Tests: t-tests (one-sample, independent two sample, paired), Chi-square test, One-Way and Two-Way ANOVA; Applications: Testing differences in gene expression between treatments, patient groups, or populations.	08
IV	**ADVANCED DATA ANALYSIS TECHNIQUES:** Regression Analysis: Simple linear regression, Multiple regression; Survival Analysis: Kaplan-Meier curves, Cox Proportional-Hazards model; Multivariate Analysis: Principal Component Analysis (PCA), Cluster Analysis; Applications: Gene expression predictions using regression, survival analysis in clinical trials, and clustering genes with similar expression patterns.	08

V	**STATISTICAL SOFTWARE AND DATA VISUALIZATION:** Statistical Software: Introduction to R, SPSS, and Python for bioinformatics data analysis; Data Visualization: Creating effective plots (histograms, box plots, scatter plots) and interpreting results; Application: Visualizing and interpreting gene expression data, clinical trial results.	08

TEXT BOOKS / REFERENCES

1. Wayne W. Daniel, Chad L. Cross, "Biostatistics: A Foundation for Analysis in the Health Sciences", 10th edition, Wiley, 2013.
2. S. N. Gupta, "Fundamentals of Biostatistics", 5th edition, Himalaya Publishing House, 2015.
3. Dr. B. K. Mahajan, "Biostatistics and Research Methodology", 1st edition, Jaypee Brothers Medical Publishers, 2016.
4. Dr. S. K. Srivastava, A. Bhardwaj, "Biostatistics – A Practical Approach", 1st edition, Global Vision Publishing House, 2010.
5. Marcello Pagano, Kimberlee Gauvreau, "Principles of Biostatistics", 2nd edition, Brooks/Cole, 2000.

BBI651 IMMUNOINFORMATICS LAB		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Identify targets and retrieve peptide/protein sequences for immunoinformatics analysis.	**K2**
CO2	Predict MHC Class I binding peptides and proteasomal cleavage sites using online tools (SYFPEITHI, BIMAS, PAProC, NetChop).	**K3**
CO3	Predict MHC Class II binding peptides and examine their immunological relevance.	**K3**
CO4	Predict B-cell epitopes (sequential and conformational) using bioinformatics tools.	**K3**
CO5	Apply antigen–antibody interaction studies and interpret immunoinformatics results for vaccine/therapeutic design.	**K3**

LIST OF EXPERIMENTS
1. Identification of Target. **2.** Prediction of MHC-I binding peptides using SYFPEITHI database. **3.** Prediction of MHC-I binding peptides using BIMAS database. **4.** Proteasomal cleavage prediction using PAProC. **5.** Proteasomal cleavage prediction using NetChop. **6.** Prediction of MHC-II binding peptides. **7.** Prediction of B-Cell Sequential epitopes. **8.** Prediction of B-Cell Conformational epitopes. **9.** Antigen-Antibody Interaction study.
Note: The instructor may add / delete / modify / tune experiments as required.

BBI652 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING IN BIOINFORMATICS LAB		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Demonstrate the use of Python and AI tools for data analysis.	**K2**
CO2	Perform data pre-processing and visualization using machine learning libraries.	**K3**
CO3	Apply supervised machine learning algorithms to biological datasets.	**K3**
CO4	Apply unsupervised machine learning techniques for data clustering and analysis.	**K3**
CO5	Apply evaluation metrics to interpret machine learning results for bioinformatics applications.	**K3**

LIST OF EXPERIMENTS

1. Introduction to Python programming and Jupyter Notebook/Google Colab.
2. Study of Python libraries: NumPy, Pandas, Matplotlib, and Scikit-learn.
3. Loading, exploring, and visualizing biological datasets using Pandas and Matplotlib.
4. Data pre-processing: handling missing values, feature scaling, and splitting data into training and testing sets.
5. Implementation of Linear Regression and Logistic Regression using Scikit-learn.
6. Implementation of Decision Tree and Random Forest for classification.
7. Implementation of K-Nearest Neighbour (KNN) and Support Vector Machine (SVM) for classification.
8. Implementation of K-Means Clustering and visualization of clustered data.
9. Model performance evaluation using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.
10. Case study: Disease prediction using a healthcare dataset.
11. Case study: Gene expression or DNA sequence classification using a publicly available bioinformatics dataset.

Note: The instructor may add / delete / modify / tune experiments as required.

BBI653 JAVA PROGRAMMING LAB		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO1	Implement basic data structures using arrays and string-handling techniques.	**K3**
CO2	Apply core OOP concepts like inheritance and polymorphism for modular code reuse.	**K3**
CO3	Construct organized code architectures using packages and abstract interfaces.	**K3**
CO4	Build robust applications by managing runtime errors with exception handling and I/O streams.	**K3**
CO5	Develop interactive, event-driven user interfaces using AWT, event models, and applets.	**K3**

LIST OF EXPERIMENTS
1. Programs to implement the following Java Programming Techniques. **2.** Arrays. **3.** Inheritance. **4.** Polymorphism. **5.** Package and Interface. **6.** Exception Handling. **7.** String handling. **8.** AWT, Event Handling & I/O. **9.** Applets.
Note: The instructor may add / delete / modify / tune experiments as required.